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Inhalation Device Including Substance Usage Control

Abstract

An apparatus including a first cartridge, a sensor, and a controller. The first cartridge can include a first release device configured to release a first substance into a housing. The controller can be configured to receive data from the sensor. The controller can determine an amount of first substance released by the first cartridge based on the data. The first release device can be controlled based on the determined amount of first substance.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/541,677, filed Dec. 3, 2021, which is a continuation of U.S. patent application Ser. No. 16/288,890, filed Feb. 28, 2019, which is a continuation of U.S. patent application Ser. No. 14/257,079, filed Apr. 21, 2014, now U.S. Pat. No. 10,420,373, which is a continuation of U.S. patent application Ser. No. 13/460,982, filed May 1, 2012, now U.S. Pat. No. 8,714,150, which is a continuation of U.S. patent application Ser. No. 13/215,728, filed Aug. 23, 2011, now U.S. Pat. No. 8,550,069, which claims the benefit of the filing date of U.S. Provisional Application No. 61/402,136, filed Aug. 24, 2010, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] The present disclosure relates generally to the field of smoke inhalation devices and more particularly to control mechanisms for smoke inhalation devices.

[0003] Smoke inhalation devices, commonly known as e-cigarettes, can be used to simulate a cigarette or a cigar. For example, a smoke inhalation device can vaporize a liquid including nicotine. A user of the smoke inhalation device can inhale the vapor and have an experience similar to smoking a traditional cigarette or cigar.

SUMMARY

[0004] One illustrative embodiment is related to an apparatus including a first cartridge, a sensor, and a controller. The first cartridge can include a first release device configured to release a first substance into a housing. The controller can be configured to receive data from the sensor. The controller can determine an amount of first substance released by the first cartridge based on the data. The first release device can be controlled based on the determined amount of first substance.

[0005] Another illustrative embodiment is related to a method for metering an inhalation device. The method can include receiving data from a sensor. An amount of a first substance released by a first cartridge can be determined based on the data. The first cartridge can include a first release device configured to release the first substance into a housing. The first release device can be controlled based on the determined amount of first substance.

[0006] Another illustrative embodiment is related to a non-transitory computer-readable medium having instructions stored thereon that, if executed by a computing device, cause the computing

device to perform operations for metering an inhalation device. The method can include receiving data from a sensor. An amount of a first substance released by a first cartridge can be determined based on the data. The first cartridge can include a first release device configured to release the first substance into a housing. The first release device can be controlled based on the determined amount of first substance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several embodiments in accordance with the disclosure and are, therefore, not to be considered limiting of its scope, the disclosure will be described with additional specificity and detail through use of the accompanying drawings.

[0008] FIG. 1 is a section view of an inhalation device in accordance with an illustrative embodiment.

[0009] FIG. 2 is a schematic of an inhalation device control system in accordance with an illustrative embodiment.

[0010] FIG. 3 is a schematic of the inhalation device control system of FIG. 2 with additional sensors in accordance with an illustrative embodiment.

[0011] FIG. 4 is a diagram of a first inhalation device in accordance with an illustrative embodiment.

[0012] FIG. 5 is a diagram of a second inhalation device in accordance with an illustrative embodiment.

[0013] FIG. 6 is a diagram of a third inhalation device in accordance with an illustrative embodiment.

[0014] FIG. 7 is a diagram of a fourth inhalation device in accordance with an illustrative embodiment.

[0015] FIG. 8 is a diagram of a fifth inhalation device in accordance with an illustrative embodiment.

[0016] FIG. 9 is a diagram of a sixth inhalation device in accordance with an illustrative embodiment.

DETAILED DESCRIPTION OF THE ILLUSTRATIVE EMBODIMENTS

[0017] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and make part of this disclosure.

[0018] The present disclosure is directed to an inhalation device including a substance usage control mechanism. The inhalation device can comprise a housing to hold and contain parts, at least one cartridge containing a substance such as nicotine, at least one releasing device such as a heater or an atomizer to work with the cartridge, a power source such as a battery, and a controller. The controller can collect data and control the cartridge and the releasing device to deliver a predetermined amount of the substance to the user. The inhalation device can be a long cylindrical shape similar to a cigarette or cigar. Advantageously, the inhalation device can control and/or limit

amount of substance delivered the usage to help the user limit the consumption of a certain substance such as nicotine. Advantageously, the inhalation device can control and/or limit amount of substance delivered such that a user can set a limit on an intake amount of the substance per use or for a certain period by the user. Advantageously, the inhalation device can receive and transmit data related to settings and usage. Advantageously, the inhalation device can monitor an amount of substance delivered to a user.

[0019] Referring to FIG. 1, a section view of an inhalation device **100** in accordance with an illustrative embodiment is shown. The inhalation device **100** can include a housing **112**, a first cartridge **114**, a second cartridge **116**, a power source **122**, a controller **124**, a communication device **146**, and a light device **126**. In other embodiments, one or any number of cartridges can be used.

[0020] The housing **112** can be a long cylindrical shape, for example, the shape and size of a cigar, cigarillo, or cigarette. In other embodiments, the inhalation device **100** can be configured as other smoking or vapor delivery articles such as a hookah, a pipe, an inhaler, or a humidifier. The housing **112** can include an outlet hole **128** at a first end (i.e., a mouth end), where a user can place his lips to breathe in vapor or gas generated by inhalation device **100**. The housing **112** can include an inlet hole **130** to allow air to enter the housing **112** when a user sucks on the first end. The housing **112** can be made of paper (e.g., paper rolled up in tubular shape), plastic, metal, wood, glass, or any other material.

[0021] The first cartridge **114** can be configured to retain and release a first substance such as nicotine. For example, the first cartridge **114** can include a medium such as a cotton fabric or a comparable material rolled up or shaped to fit inside housing **112**. The cotton fabric can be soaked with the first substance. The first substance can include any drug, scent, transport agent, coloring, or a combination thereof. In one embodiment, the first substance is configured such that the substance can be vaporized or gasified. The first cartridge **114** can include a first release device **118**. The first release device **118** can be embedded in or placed near the medium of the first cartridge **114**. The first release device **118** can include one or a combination of a heating element, a vibration generator such as an ultrasonic or a piezoelectric transducer, an atomizer, or any other actuator to energize particles of the first substance such that the particles are released from the first cartridge **114** into a gas form or microscopic droplets that can be carried off by air stream, for example, an airstream between the inlet hole **130** and outlet hole **128**. The first release device **118** can be configured to control or meter the amount or rate of first substance released from the first cartridge **114**. In some embodiments, the inhalation device includes a single cartridge.

[0022] The second cartridge **116** can be configured to retain and release a second substance such as a scent. For example, the second cartridge **116** can include a medium such as a cotton fabric or a comparable material rolled up or shaped to fit inside housing **112**. The cotton fabric can be soaked with the second substance. The second substance can include any drug, scent, transport agent, coloring, or a combination thereof. In one embodiment, the first substance and the second substance are difference substances. In another embodiment, the first substance and the second substance are the same substance. In one embodiment, the second substance is configured such that the substance can be vaporized or gasified. The second cartridge **116** can include a second release device **120**. The second release device **120** can be embedded in or placed near the medium of the second cartridge **116**. The second release device **120** can include one or a combination of a heating element, a vibration generator such as an ultrasonic or a piezoelectric transducer, an atomizer, or any other actuator to energize particles of the second substance such that the particles are released from the second cartridge **116** into a gas form or microscopic droplets that can be carried off by airstream, for example, an airstream between the inlet hole **130** and outlet hole **128**. The second release device **120** can be configured to control or meter the amount or rate of second substance released from the second cartridge **116**. The first cartridge **114** and the second cartridge **116** can be arranged such that the airstream passes over and/or through both the first cartridge **114** and the

second cartridge **116**. In other embodiments, the inhalation device can include multiple cartridges. [0023] The power source **122** can provide power to the first cartridge **114**, first release device **118**, the second cartridge **116**, second release device **120**, the controller **124**, the communication device **146**, and the light device **126**. The power source **122** can be a battery such as an alkaline battery, a nickel metal hydride battery, a lithium-ion battery, a rechargeable battery or any other energy source. In other embodiments, the power source **122** can be external to the housing **112**.

[0024] The light device **126** is configured to simulate the burning on the tip of a cigar or cigarette. The light device **126** can include a light emitting diode (LED), a resistance-type lamp, an organic light emitting diode (OLED), or any other light emitting device. The light device **126** can be configured to vary intensity based on the amount of airflow between the inlet hole **130** and outlet hole **128**.

[0025] The controller **124** can be configured to control the first release device **118**, the second release device **120**, the communication device **146**, the power source **122**, and the light device **126**. The controller **124** can be operatively and communicatively coupled to the first release device **118**, the second release device **120**, the communication device **146**, and the light device **126**. The controller **124** can be a processor such as a central processing unit, an application specific integrated circuit (ASIC), a state machine or any other controller. The controller **124** can include memory which can be any type of permanent or removable computer memory known to those of skill in the art. The memory of controller **124** can be a computer-readable storage medium. The controller **124** can include software for controlling the first release device **118**, the second release device **120**, the communication device **146**, and the light device **126**, which can be implemented as computer-readable instructions configured to be stored on the memory of controller **124**.

[0026] The communication device **146** can be configured to allow a user to change and monitor the settings and state of the inhalation device **100**. For example, in one embodiment, communication device **146** can be used to reprogram a part of the control logic of controller **124** to limit the usage of the inhalation device **100**. The communication device **146** can include at least one of a switch, a keypad, a display, an input/output port, and a wireless transceiver. In one embodiment, the input/output port and the wireless transceiver can be employed to create a communications link between the controller **124** and an external computer, such as a cell phone or personal computer.

[0027] Referring to FIG. 2, a schematic of an inhalation device control system **200** in accordance with an illustrative embodiment is shown. The inhalation device control system **200** can include a first release device **218** associated with a first cartridge **214**, a second release device **220** associated with a second cartridge **216**, a power source **222**, a controller **224**, a timer **248**, a counter **242**, sensors **240**, a communication device **246**, and a light device **226**. The controller **224** can be operatively and communicatively coupled to the first release device **218**, the second release device **220**, the power source **222**, sensors **240**, the communication device **246**, and the light device **226**. The inhalation device control system **200** can be, for example, discrete, integrated as a chip, or a printed circuit board assembly.

[0028] The first cartridge **214** and second cartridge **216** may each contain a unique substance or a mixture of two or more substances and the mixture composition for one cartridge is different from another. For this reason, the first release device **218** and the second release device **220** can each be operated in a unique way to optimize the effect of the substance for the user. Furthermore, different techniques can be employed for releasing the substances of different cartridges.

[0029] Sensors **240** can be, for example, a flow sensor, a thermocouple or pressure sensor to detect the air stream passing through the device or a group of sensors to detect any one of or a combination of airflow, temperature change, pressure change, illumination of light, current change, voltage change, and other physical and electrical properties. Sensors **240** can be located within or on a housing associated with the inhalation device control system **200**.

[0030] The controller **224** can be configured to control and monitor the first release device **218**, the second release device **220**, sensors **240**, the communication device **246**, the power source **222**, and

the light device **226**. The controller **224** can be a processor such as a central processing unit, an application specific integrated circuit (ASIC), a state machine or any other controller. The controller **224** can include a memory **280** which can be any type of permanent or removable computer memory known to those of skill in the art. The memory **280** of controller **224** can be a computer-readable storage medium. The controller **224** can include software for controlling the first release device **218**, the second release device **220**, the communication device **246**, and the light device **226**, which can be implemented as computer-readable instructions configured to be stored on the memory **280** of controller **224**.

[0031] The communication device **246** can include at least one of a switch **271**, a keypad **273**, a display **250**, an input/output port **277**, and a wireless transceiver **279**. The input/output port **277** can be, for example, a serial port. The wireless transceiver **279** can be, for example, a Wi-Fi transceiver compliant with the IEEE 802.11 standard or a Bluetooth transceiver.

[0032] In one embodiment, the switch **271** can be a simple manual switch on a housing for sending reprogramming instructions to the controller **224** by clicking the switch **271** (e.g., clicking the switch five times with the clicking speed at one click per second or faster can reprogram the controller **224** to allow five puffs in a given period and then turn the device off). In another embodiment, the switch **271** can be a dial located on or in the middle of the housing for sending reprogramming instructions to the controller **224** such that by turning the dial the user can set the limit amount of the substance to be taken or inhaled. The dial switch can also be a section of the housing itself. In another embodiment, the switch **271** can be a shock or sound sensor such as a microphone for the user to tap for sending reprogramming instructions to the controller **224**. The user can give the device taps at certain speed to set a desired amount of the substance to intake or inhale in a given period. In another embodiment, the switch **271** can be a light sensor to detect energy emitted by a laser, infrared, or other visible or invisible light source. Reprogramming instructions to the controller **224** can be on or off signals from the light source.

[0033] The user may use communication device **246** to reprogram the limit value in the controller **224**. For example, a simple form of communication device **246** can be a manual switch the user can press or turn on or around the housing. By clicking or turning the switch manually, the user can change the limit setting in the controller **224**. For example, by clicking the switch four times with a frequency of 1 Hz or higher, the user can smoke one cigarette equivalent worth of substance in each cycle or 10 each time, and clicking the switch two times will give a half cigarette worth of substance in each cycle. If the manual switch is a dial switch on the housing, the dial setting can set the limit. For example, the dial has indicator lines around it on the non-moving portion of the housing, and setting the dial at one can be equivalent to one cigarette dose per cycle and setting it to one half means a half of a cigarette dose.

[0034] The display **250** can be located on within or on the housing associated with the inhalation device control system **200**. The display **250** can show messages indicating the usage to help the user keep track of his use of the device. The display **250** can be, for example, a light, a series of lights, a segmented display or a pixelated display. The display **250** can be made, for example, of LEDs, a LED panel, a liquid crystal display, a group of small light bulbs, or electronic ink. The display **250** can display indications to the user, for example, as letters, dots, symbols, a series of lines, figures, pictures, shapes or signals. The display **250** can be monotone or multi-colored.

[0035] The display **250** can indicate the usage level for the user to monitor the use of the device and can send out messages to encourage the user to cut down on the use based on the progress he or she makes. A display message can include a number of cigarette equivalents used or left on the inhalation device control system **200**, a number of puffs (i.e., drags) used or left on the inhalation device control system **200**, and it can include a real time display of a number of puffs the user is allowed to take in each cycle as he or she uses the inhalation device control system **200**. This way the device alerts the user, as he or she uses it, how many cigarette equivalents he or she is taking in a given period since using the same device for a long time may lead the user to a false sense of the

substance usage. For example, the inhalation device control system **200** with a full first cartridge **214** and second cartridge **216** can be equivalent to one pack of cigarettes and the display **250** can clearly indicate the number of cigarette equivalents or puffs left in the first cartridge **214** and second cartridge **216** as he or she uses the inhalation device control system **200**.

[0036] The timer **248** can be configured to measure an amount of the substance disbursed and/or to meter a predetermined amount of the substance to a user. The timer **248** can be separate from or integrated with controller **224**. The timer **248** can be configured to measure, for example, the amount of time elapsed between user puffs, the amount of time the first release device **218** and the second release device **220** are activated, and the time between uses.

[0037] The timer **248** can be configured to measure an amount of the substance disbursed and/or to meter a predetermined amount of the substance to a user. The timer **248** can be separate from or integrated with controller **224**. The timer **248** can be configured to measure, for example, the amount of time elapsed between user puffs, the amount of time the first release device **218** and the second release device **220** are activated, and the time between uses. The timer **248** can be activated, for example, by an airflow sensor or a pressure switch. The timer **248** can be configured to measure a duration of a puff while the sensors **240** are activated.

[0038] The counter **242** can be configured to count a number of puffs taken per given period by a user. The counter **242** can be separate from or integrated with controller **224**. The counter **242** can be configured to measure, for example, a number of puffs taken by a user during a smoking session, a day, a week, a month, or any other time period. The counter **242** can be activated, for example, by sensors **240**.

[0039] For a more precise control of the amount of the substance allowed in a given period, timer **248** can be used to time the airflow through the device. Each time sensors **240** such as a flow sensor or a pressure switch are activated by the airflow, timer **248** can measure the duration of the airflow and the time for duration can be accumulated in a given period to activate a limit feature according to a substance distribution limit set in the control logic. Also counter **242** can be used to count the number of activations of sensors **240**.

[0040] Counter **242** and timer **248** can provide usage data on the number of puffs per given period and timer **248** can provide both time duration of airflow for each puff and total time accumulated for the airflow over a given period. Therefore, when the substance distribution limit is set by the user or preset by the factory, the limit can be set on the actual amount of the substance in terms of total time allowed for airflow within a given period for use, not just an allowable number of puffs. Since an amount of airflow for a puff is different from person to person, setting a number of puffs as a limit can allow different amount of the targeted substance delivered to each different user or even to a same user depending on duration of each and every puff. Therefore, using timer **248** to monitor the usage along with counter **242** can provide a more accurate estimate of the measurement of the total airflow.

[0041] The inhalation device control system **200** can come with a factory setting for the allowable amount of the substance contained in the cartridges (**214**, **216**) to be released. Different users have different usage habits: One person may prefer long hard puffs and another person may like short shallow puffs. A person having long hard puffs usually finishes a cigarette in a less number of puffs than the person having the short shallow puffs. For this reason, the amount of substance allowed can be set as total duration time that is a time accumulation value of the total puffs in a given period. Another sensor such as a current sensor for the releasing means or the power source or a flow sensor with the air speed measurement can be applied to estimate the quantity of the substance use whenever the user draws the air through the device, and this information can be used along with the timer to calculate the more accurate amount of the substance use. Once the limit value as the time accumulation is set, timer **248** measures the duration of each puff and send the data to the control logic in controller **224** and controller **224** accumulates and compares the accumulated data with the set limit value. Once the accumulated time value or the value calculated from a

combination with the accumulated time and the data of the quantity of the substance from the sensor reaches or goes over the limit value, controller **224** turns the inhalation device control system **200** off or makes the inhalation device control system **200** go into a low level mode in which the inhalation device control system **200** releases the substance at a lower rate than during the earlier period. The limit value can be set with cycles. For example, once the device is on, the user can be allowed a first set time limit for the total duration time of puffs within five minutes that is roughly equal to smoking one cigarette, and within a given period such as twelve hours or a day, the user can be allowed to use the inhalation device control system **200** a set number of times. Once the limit is reached controller **224** can turn off the releasing means (**218**, **220**) to stop the substance flow to the user until the next cycle begins after non smoking time expires between two consecutive cycles or controls the releasing means (**218**, **220**) such that the substance flow to the user is more limited per puff until the next cycle begins.

[0042] Referring to FIG. 3, a schematic of the inhalation device control system **200** of FIG. 2 with additional sensors in accordance with an illustrative embodiment is shown. The inhalation device control system **200** can include a first release device **218** associated with a first cartridge **214**, a second release device **220** associated with a second cartridge **216**, a power source **222**, a controller **224**, a timer **248**, a counter **242**, sensors **240**, a communication device **246**, and a light device **226**. The controller **224** can be operatively and communicatively coupled to the first release device **218**, the second release device **220**, the power source **222**, sensors **240**, the communication device **246**, and the light device **226**. The communication device **246** can include at least one of a switch **271**, a keypad **273**, a display **250**, an input/output port **277**, and a wireless transceiver **279**. As discussed further below, the inhalation device control system **200** can measure substance usage in ways other than using a flow sensor or a pressure switch.

[0043] The first release device **218** and the second release device **220** can release a first substance from the first cartridge **214** and a second substance from the second cartridge **216** using, for example, heat and/or vibration. A temperature sensor **291** located on the second cartridge **216** can be used to detect the temperature of the second cartridge **216**. The controller **224** can estimate an amount of substance released by the second cartridge **216** based on temperature information from the temperature sensor **291**, along with information from timer **248**. A vibration sensor **292** located on the second cartridge **216** can be used to detect the agitation of the second cartridge **216**. The controller **224** can estimate an amount of substance released by the second cartridge **216** based on agitation information from the vibration sensor **292**, along with information from timer **248**. A current sensor **293** located on the power feed to the first release device **218** can be used to determine the current used by the first release device **218**. The controller **224** can estimate an amount of substance released by the first cartridge **214** based on current information from the current sensor **293**. Alternatively, a voltage sensor can be used.

[0044] In another embodiment, a current sensor **294** located on the power feed from power source **222** can be used to estimate the current used by the first release device **218** and second release device **220**. The controller **224** can estimate an amount of substance released by the first cartridge **214** and the second cartridge **216** based on current information from the current sensor **294**.

Alternatively, a voltage sensor can be used. In another embodiment, a light sensor **295** located in proximity to light device **226** can be used to estimate the usage of the first release device **218** and second release device **220**. The controller **224** can estimate an amount of substance released by the first cartridge **214** and the second cartridge **216** based on the amount of time the light sensor **295** is activated as well as the intensity detected by the light sensor **295**. Alternatively, other sensors can be used as is known in the art. Alternatively, sensors **291-295** can be placed in various locations as is known in the art.

[0045] An illustrative inhalation device package for the market can be, for example, a smoking device in a cigarette form that comes with a capacity for smoking about twenty cigarettes. A factory setting can be preloaded to the smoking device to allow about fifteen puffs per use within five

minutes, and after the user is done smoking for the time, he or she has to wait at least 30 minutes before using the device again. A display on the inhalation device can show the number of cigarette equivalents left to go, and it can also show the user the number of puffs left to go before the current use is expired as he or she use the inhalation device. In such a setting, the inhalation device can be programmed to monitor the usage of the substance contained in a cartridge by measuring the number of intake, or calculating the quantity of consumption of the substance from data collected from timer for the duration of each and every intake of the air and another sensor such as a electric current sensor or air flow sensor for the rate of flow of the substance. The number of puffs can then be translated to actual substance used based on the calculated estimation from the data provided by the timer and the sensor. Also a counter in the system can be used to count the number of intakes to provide the data to the control means to make calculation simpler. The user may use a button or a dial to change the factory settings such as the number of puffs allowed per use (e.g., from fifteen puffs setting to twenty puffs) and the duration of off time after the use before the next use is permitted (e.g., 30 minutes factory setting to 45 minutes or to 15 minutes). There can be one or more buttons or dials on the inhalation device to reprogram the settings for the inhalation device. A first button or dial for setting the off time duration and a second button or dial for setting a number of puffs allowed per use or other settings depending on the user preference and convenience.

[0046] FIGS. **4** through **8** depict a number of different illustrative embodiments of the inhalation device. Referring to FIG. **4**, a diagram of a first inhalation device **400** in accordance with an illustrative embodiment is shown. The first inhalation device **400** can include a housing **412**, a communication device **446**, and a light device **426**. The communication device **446** can include a display **450** and buttons **471** and **472**. The first inhalation device **400** has two separate programming modes. In a first mode, button **471** can be used to change the limit for a number of puffs allowed per use. In a second mode, button **472** can be used to change the amount of shut off time required before the next use. The display **450** can display an indication of a setting. (e.g., “10,” “30,” “60,” “120,” “10,” “15,” “20,” “25”)

[0047] Referring to FIG. **5**, a diagram of a second inhalation device **500** in accordance with an illustrative embodiment is shown. The second inhalation device **500** can include a housing **512**, a communication device **546**, and a light device **526**. The communication device **546** can include a first display **550**, a button **571**, and a second display **551**. The button **571** can be used to change the amount of shut off time required before the next use. The first display **550** can display an indication of a setting. (e.g., “10,” “30,” “60,” “90”) The second display **551** can include lights **552**. The lights **552** can indicate the indicating the number of cigarette equivalent remaining in the second inhalation device **500**.

[0048] Referring to FIG. **6**, a diagram of a third inhalation device **600** in accordance with an illustrative embodiment is shown. The third inhalation device **600** can include a housing **612**, a communication device **646**, and a light device **626**. The communication device **646** can include a first display **650**, a button **671**, and a second display **651**. The button **671** can be used to change the amount of shut off time required before the next use. The first display **650** can display an indication of a setting. (e.g., “10,” “30,” “60,” “90”) The second display **651** can display an indication (e.g., “385”) showing the number of puffs or substance quantity remaining or available in the second inhalation device **600**.

[0049] Referring to FIG. **7**, a diagram of a fourth inhalation device **700** in accordance with an illustrative embodiment is shown. The fourth inhalation device **700** can include a housing **712**, a communication device **746**, and a light device **726**. The communication device **746** can include a first display **750** and a dial **771**. The dial **771** can be implemented as part of the housing **712**. For example, a bezel of the dial **771** can be located at a filter area of the housing **712**. The housing **712** can include graduation markings (e.g., “10,” “30”) near the dial **771**. The dial **771** can be used to change the amount of shut off time required before the next use. The first display **750** can display an indication (e.g., “20”) showing the number of cigarette equivalents remaining or available in the

fourth inhalation device **700**.

[0050] Referring to FIG. **8**, a diagram of a fifth inhalation device **800** in accordance with an illustrative embodiment is shown. The fifth inhalation device **800** can include a housing **812**, a communication device **846**, and a light device **826**. The communication device **846** can include a first display **850**, a button **871**, and a second display **851**. The button **871** can be used to change the usage limit settings by the user. The first display **850** can display an indication of a setting. (e.g., “10,” “30,” “60,” “90”) The second display **851** can display an indication (e.g., “20”) showing the number of cigarette equivalents remaining or available in the fifth inhalation device **800**.

[0051] Referring to FIG. **9**, a diagram of a sixth inhalation device **900** in accordance with an illustrative embodiment is shown. The sixth inhalation device **900** can include a housing **912**, a communication device **946**, and a light device **926**. The communication device **946** can include a first display **950**, buttons **971** and **972**, and a second display **951**. The sixth inhalation device **900** has two separate programming modes. In a first mode, button **971** can be used to change the limit for a number of puffs allowed per use. In a second mode, button **972** can be used to change the amount of shut off time required before the next use. The first display **950** can display an indication of a setting. (e.g., “10,” “30,” “60,” “120,” “10,” “15,” “20,” “25”) The second display **951** can display an indication (e.g., “20”) showing the number of cigarette equivalents remaining or available in the sixth inhalation device **900**.

[0052] One or more flow diagrams may have been used herein. The use of flow diagrams is not meant to be limiting with respect to the order of operations performed. The herein described subject matter sometimes illustrates different components contained within, or connected with, different other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected”, or “operably coupled”, to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable”, to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

[0053] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

[0054] It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to inventions containing only one such recitation, even when the same claim includes the introductory

phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

[0055] The foregoing description of illustrative embodiments has been presented for purposes of illustration and of description. It is not intended to be exhaustive or limiting with respect to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from practice of the disclosed embodiments. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents.

Claims

1. An inhalation device comprising: a controller configured to control operation of a heater; wherein the controller is configured to turn off the heater when a measured duration of airflow through the inhalation device for each puff reaches or exceeds a limit value.
2. The inhalation device according to claim 1, further comprising a timer configured to measure the duration of airflow through the inhalation device for each puff.
3. The inhalation device according to claim 2, wherein the timer is separate from the controller.
4. The inhalation device according to claim 2, wherein the timer is integrated with the controller.
5. The inhalation device according to claim 1, further comprising a sensor configured to detect one or more of an airflow, a temperature change, a pressure change, illumination of light, a current change, and a voltage change.
6. The inhalation device according to claim 5, wherein the sensor is at least one of an airflow sensor, a thermocouple, a pressure sensor, a shock sensor, a sound sensor, or a microphone.
7. The inhalation device according to claim 2, further comprising a sensor configured to detect one or more of an airflow, a temperature change, a pressure change, illumination of light, a current change, and a voltage change, wherein the timer is configured to be activated by the sensor.
8. The inhalation device according to claim 2, further comprising a sensor configured to detect one or more of an airflow, a temperature change, a pressure change, illumination of light, a current change, and a voltage change, wherein the timer measures a puff duration while the sensor is activated.
9. The inhalation device according to claim 1, wherein the limit value is set as a total time allowed for airflow through the inhalation device within a given period of use.
10. The inhalation device according to claim 1, further comprising a cartridge that contains a

substance configured to be released by the heater.

11. The inhalation device according to claim 1, further comprising a cartridge including a heater and a substance configured to be released by the heater.

12. The inhalation device according to claim 1, wherein the controller is configured to prevent turning on the heater during a shut off time following the measured duration of airflow reaching or exceeding the limit value.

13. An inhalation device comprising: a controller configured to control operation of a heater; wherein the controller is configured to turn off the heater when a timed airflow through the inhalation device that passes through an inlet hole and an outlet hole of the device reaches or exceeds a limit value.

14. The inhalation device according to claim 13, further comprising a timer configured to time the airflow through the inhalation device that passes through an inlet hole and an outlet hole of the device, wherein the timer is separate from the controller.

15. The inhalation device according to claim 13, further comprising a timer configured to time the airflow through the inhalation device that passes through an inlet hole and an outlet hole of the device, wherein the timer is integrated with the controller.

16. The inhalation device according to claim 13, further comprising a sensor configured to detect one or more of an airflow, a temperature change, a pressure change, illumination of light, a current change, and a voltage change.

17. The inhalation device according to claim 13, further comprising a sensor configured to detect one or more of an airflow, a temperature change, a pressure change, illumination of light, a current change, and a voltage change, and further comprising a timer configured to time the airflow through the inhalation device that passes through an inlet hole and an outlet hole of the device, wherein the timer is configured to be activated by the sensor.

18. The inhalation device according to claim 13, further comprising a cartridge that contains a substance configured to be released by the heater.

19. The inhalation device according to claim 13, further comprising a cartridge including a heater and a substance configured to be released by the heater.

20. An inhalation device comprising: a controller configured to control operation of a heater; wherein the controller is configured to turn off the heater when an amount of time the heater is activated reaches or exceeds a limit value.
